

Project Report: Feature Engineering & Model Comparison (Task 2)

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Project Title: California Housing Price Prediction System

1. Project Objective

The objective of this task is to advance from a basic single-model setup to an industry-standard Machine Learning pipeline. This project focuses on implementing data preprocessing (Feature Scaling), training multiple regression algorithms side-by-side, and objectively selecting the best-performing model based on measurable evaluation metrics.

2. Methodology & Implementation Steps

- **Data Loading:** The California Housing Dataset was imported natively via scikit-learn , combining attributes like income, house age, population, and location parameters into a single working structure.
- **Feature Scaling:** Real-world data exists on varying numeric scales. To prevent high-magnitude features from dominating weights incorrectly, StandardScaler was applied to center data around a common scale.
- **Data Partitioning:** The dataset was split into an 80% training set and a 20% testing set to ensure proper validation against completely unseen records.
- **Multi-Model Execution:** Three distinct regression frameworks were trained : Linear Regression (Baseline) , Ridge Regression (L2 Regularization to control overfitting) , and a Decision Tree Regressor (to handle non-linear boundaries).

3. Model Performance Comparison Matrix (Deliverable 2)

The models were evaluated against the test partition, yielding the following mathematical results:

Trained Algorithm	RMSE (Lower is Better)	R ² Score (Higher is Better)	Performance Behavior
Linear Regression	~0.7455	~0.5755	Standard linear baseline prediction.

Trained Algorithm	RMSE (Lower is Better)	R ² Score (Higher is Better)	Performance Behavior
Ridge Regression	~0.7455	~0.5755	L2 Regularization; stable weights.
Decision Tree (Depth=5)	~0.7360	~0.5860	Best Performance. Captures non-linearities.

4. Key Insights & Conclusion

- **The Power of Non-Linearity:** The Decision Tree Regressor generated a higher R^2 score (~58.6%) and lower RMSE compared to linear variants. This confirms that geographic and economic housing data displays intricate non-linear patterns that standard linear models fail to absorb cleanly.
- **Preprocessing Necessity:** Feature Scaling successfully stabilized variance and provided uniform importance across all predictors during the training steps.
- **Target Capping Constraint:** Predictive performance remains restricted due to the dataset's artificial \$500k valuation cap, which introduces an aggressive non-linear roof that standard regression algorithms struggle to resolve perfectly.

Final Selection: The **Decision Tree Regressor** is chosen as the final model for this deployment phase due to its empirical superior accuracy and explanatory capabilities.